MATHEMATICS 2, PART 4, HW 2

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standard form $\xrightarrow{*}$ dual form

$$\begin{aligned} \max\ & c^Tx \\ & Ax \le b \\ & x \ge 0 \end{aligned} \quad (P) \qquad \begin{aligned} \min\ & b^Ty \\ & A^Ty \ge c \\ & y \ge 0 \end{aligned} \quad (D)$$

1. (P) dual of (D)?

rewrite

$$\begin{aligned} \max\ & (-b)^Ty \\ & (-A^T)y \le -c \\ & y \ge 0 \end{aligned}$$

dualize like *

$$\begin{aligned} \min\ & -c^Tx \\ & -Ax \ge -b \\ & x \ge 0 \end{aligned}$$

turn to max, simplify

$$\begin{aligned} \max\ & c^Tx \\ & Ax \le b \\ & x \ge 0 \end{aligned}$$ exactly (P)

2. $$\begin{aligned} \min\ & c^Tx \\ & Ax = b \\ & x \ge 0 \end{aligned} \quad (1) \qquad \begin{aligned} \max\ & b^Ty \\ & A^Ty + s = c \\ & s \ge 0,\ y \text{ unconstrained} \end{aligned} \quad (2)$$

rewrite (1) in standard form and dualize:

$$\begin{aligned} \max\ & (-c)^Tx \\ & Ax \le b \ \wedge\ -Ax \le -b \sim \begin{bmatrix} A \\ -A \end{bmatrix} x \le \begin{bmatrix} b \\ -b \end{bmatrix} \\ & x \ge 0 \end{aligned}$$

$$\begin{aligned} \min\ & \begin{bmatrix} b \\ -b \end{bmatrix}^T \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ & \begin{bmatrix} A \\ -A \end{bmatrix}^T \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \ge -c \\ & y_1, y_2 \ge 0 \end{aligned} \sim \begin{aligned} \min\ & b^Ty_1 - b^Ty_2 \\ & A^Ty_1 - A^Ty_2 \ge -c \\ & y_1, y_2 \ge 0 \end{aligned} \sim \begin{aligned} \min\ & -b^Ty \\ & -A^Ty \ge -c \\ & y \text{ unconstrained} \end{aligned}$$

/let $y = y_2 - y_1$ /let $s = c - A^Ty \ge 0$

$$\sim \begin{aligned} \max\ & b^Ty \\ & A^Ty + s = c \\ & s \ge 0,\ y \text{ unconstrained} \end{aligned}$$ exactly (2)

3. $$\begin{aligned} & Ax = b \\ & A^Ty + s = c \\ & x > 0,\ s > 0 \end{aligned} \quad (P_\mu)$$

next step solution: m... number of variables x_i

$$x_i s_i \approx \mu$$

$$\mu' = \left(1 - \frac{1}{6\sqrt{m}}\right)\mu$$

$$\begin{aligned} x' &= x + h \\ y' &= y + k \\ s' &= s + f \end{aligned} \quad \text{from} \quad \begin{aligned} &Ah = 0 \\ &A^T k + f = 0 \quad (S) \\ &h_i s_i + f_i x_i = \mu' - x_i s_i \;\; \forall i \end{aligned}$$

5. $\min -3x_1 - 4x_2$

$3x_1 + 3x_2 + 3x_3 = 4$

$3x_1 + x_2 + x_4 = 3$

$x_1 + 4x_2 + x_5 = 4$

$x_i \geq 0 \;\; \forall i$

$$(X) \sim \begin{aligned} &\min c^T x \\ &Ax = b \\ &x \geq 0 \end{aligned} \quad \text{, where} \quad c = \begin{bmatrix} -3 \\ -4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 3 & 3 & 0 & 0 \\ 3 & 1 & 0 & 1 & 0 \\ 1 & 4 & 0 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 3 \\ 4 \end{bmatrix}$$

- dualize

$\max b^T y$

$A^T y \leq c \sim A^T y + s = c$

$s \geq 0$, y unconstrained

- write out

$\max 4y_1 + 3y_2 + 4y_3$

$3y_1 + 3y_2 + y_3 + s_1 = -3$

$3y_1 + y_2 + 4y_3 + s_2 = -4$

$3y_1 + s_3 = 0$ $\quad (XD)$

$y_2 + s_4 = 0$

$y_3 + s_5 = 0$

$s_i \geq 0 \;\; \forall i$

6. Let $x = \left(\frac{2}{5}, \frac{8}{15}, \frac{2}{5}, \frac{19}{15}, \frac{22}{15}\right)^T$ strictly feasible?

$y = \left(-\frac{4}{5}, -\frac{4}{5}, -\frac{2}{3}\right)^T$ $\quad x_i > 0,\; s_i > 0 \;\; \forall i$

$s = \left(\frac{37}{15}, \frac{28}{15}, \frac{12}{5}, \frac{4}{5}, \frac{2}{3}\right)^T$

equalities:

(X) $\frac{60}{15} = 4$, $\frac{45}{15} = 3$, $\frac{60}{15} = 4$

(XD) $\frac{-45}{15} = -3$, $\frac{-60}{15} = -4$, $\frac{0}{5} = 0$, $\frac{0}{5} = 0$, $\frac{0}{3} = 0$ ✓

✓ ⌒

$$\boxed{\underset{\min}{c^T x} - \underset{\max}{b^T y} = x^T s \approx m\mu}$$

7. good starting point?

centrality measure $\sigma^2 = \sum_{i=1}^{m} \left(\frac{x_i s_i}{\mu} - 1\right)^2$ should be $\underline{\leq \frac{1}{36}}$ *

$\mu = \frac{x^T s}{m} = \frac{\frac{74}{15}}{5} = \frac{74}{75} \approx 0.987$ minimizes it

if we use $\mu = 1$, then $\sigma^2 = \frac{1}{405}$ which is appropriate

8. we use $\mu < 10^{-9}$ as stop condition

standard converges to $\approx -\frac{44}{9}$ in 268 iterations

9. we search the interval $[0, 1 - \frac{1}{6\sqrt{m}}]$ to find δ that satisfies * for $\mu' = \delta\mu$, for example with <u>bisection</u>

"adaptive" converges in 10 iterations

10. we get $s_1 = 0$, $s_2 = 0$, $x_3 = 0$, $s_4 = 0$, $x_5 = 0$ so x_1, x_2, x_4 are active. †

11. assume †

$$Ax = b \sim \begin{matrix} 3x_1 + 3x_2 = 4 \\ 3x_1 + x_2 + x_4 = 3 \\ x_1 + 4x_2 = 4 \end{matrix} \longrightarrow x = \left(\frac{4}{9}, \frac{8}{9}, 0, \frac{7}{9}, 0\right)$$

(active rows)

$$A^T y = c \sim \begin{matrix} 3y_1 + 3y_2 + y_3 = -3 \\ 3y_1 + y_2 + 4y_3 = -4 \\ y_2 = 0 \end{matrix} \longrightarrow y = \left(-\frac{8}{9}, 0, -\frac{1}{3}\right) \text{ and } s = \left(0, 0, \frac{8}{3}, 0, \frac{1}{3}\right)$$

feasibility holds by construction ✓

optimality? $c^T x = -3 \cdot \frac{4}{9} - 4 \cdot \frac{8}{9} = -\frac{44}{9}$ $\quad b^T y = 4 \cdot \left(-\frac{8}{9}\right) + 4 \cdot \left(-\frac{1}{3}\right) = -\frac{44}{9}$ ✓

12. HIGHS uses fewer iterations, is 5x faster (C++) and reaches exact solution (selects active + solves analytically).

Let $Ax \leq b$ and $\Phi = \{x;\ \text{feasible solution}\}$

Then $\underline{s(x) = b - Ax}$ and $I = \{ i;\ \text{exists } x \in \Phi \text{ such that } s(x)_i > 0 \}$

analytical center solution maximizes $\prod_{i \in I} s(x)_i$ constraint not active

13. $\bar{x} = \frac{1}{k} \sum_{i \in I} x^{(i)}$

1. $\bar{x}$ is feasible because Φ convex region and $\sum \frac{1}{k} = 1$
2. $s(\bar{x})_j = b_j - (A\bar{x})_j = \frac{1}{k} \sum_{i \in I} b_j - (Ax^{(i)})_j = \frac{1}{k} \sum_{i \in I} s(x^{(i)})_j > 0 \quad \forall_j$ when $i = j$

14. $\arg\max_x \prod s(x)_i = \arg\max_x \sum \ln s(x)_i = \arg\min_x \underbrace{- \sum \ln s(x)_i}$

1. each term $-\ln(\cdot)$ is convex $\rightarrow$ sum convex $| f(x)$
2. let v arbitrary direction
$$f_v'' = \sum_{i \in I} \frac{(a_i^T v)^2}{s(x)_i^2} > 0$$, therefore strictly convex

15. analytical center is unique follows from 14.
if $\min = f(x) = f(y)$ for $x \neq y$ and $z = \frac{x+y}{2}$
by strict convexity $\underline{f(z) < \frac{1}{2} f(x) + \frac{1}{2} f(y) = \min}$ contradiction

16.
$$(A)\quad \begin{aligned} 2x_1 + 1x_2 &\leq 6 \\ 6x_1 + 1x_2 &\leq 15 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \end{aligned} \quad \sim \quad (A')\quad \begin{aligned} &\max f(x) \\ &s_1 = 6 - (2x_1 + x_2) \\ &s_2 = 15 - (6x_1 + x_2) \\ &s_3 = x_1 \\ &s_4 = x_2 \\ &s_i > 0 \ \forall_i \end{aligned}$$

we solved A' with BFGS:

$x_1 = 0.69 \quad x_2 = 2.01$

$s_1 = 2.61 \quad s_2 = 8.84 \quad s_3 = 0.69 \quad s_4 = 2.01$

17. add $2x_1 + 1x_2 \leq 18$ ~ $s_5 = 18 - (2x_1 + x_2)$

$x_1 = 0.66\ (-0.03)$ $x_2 = 1.87\ (-0.14)$

redundant constraint → large slack s_5 → weak push of x^*

18. add $4x_1 + 2x_1 \leq 12$ ~ $s_5 = 2s_1$

$x_1 = 0.57\ (-0.12)$ $x_2 = 1.46\ (-0.55)$

duplicate constraint → s_1 counted twice → stronger push of x^*

Note: some calculations/derivations were done separately and only written cleanly here